What the PoC prototype is doing

All open work orders and available technicians are solved first as the nightly reference routes, then re-solved after the 16 requirement triggers.

Business objective

Show how technician routes change when operating conditions shift across execution-aware, real-time, and goal-based scheduling.

Core input files

orders_40000.csv = work orders
routes_400.csv = technicians, shifts, capacities
depots_8.csv = depot locations
accounts / goal_profiles = business priorities

What the prototype now does

Builds a visual impacted subset, re-solves after targeted event changes, and presents every scenario as a map-plus-card slide that stays fully editable in PowerPoint.

Scenario scope

4 execution-aware
7 real-time
5 goal-based

Work Orders

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Name	Address	Latitude	Longitude	AccountId	GoalProfile	WorkType	Priority	ServiceTim	Assignmer	RouteNam	Sequence	SpecialtyN	TimeWind	TimeWind	MaxViolati	Descriptio	PreferredE	SLAFlag
WO00000	Synthetic c	32.95878	-96.7244	ACC066	SLA_FIRST	HVAC	3	56				HVAC	########	########	15	HVAC worl	Depot_Ricl	1
WO00000	Synthetic c	32.92664	-96.6361	ACC075	SKILL_FIRS	General	2	24					########	########	45	General w	Depot_Gar	0
WO00000	Synthetic c	33.02374	-96.6825	ACC002	SLA_FIRST	Electrical	2	44				Electrical	########	########	15	Electrical v	Depot_Plar	1
WO00000	Synthetic c	32.89063	-96.8274	ACC116	OVERTIME	Electrical	3	39				Electrical	########	########	45	Electrical v	Depot_Dall	0
WO00000	Synthetic c	32.72715	-97.1022	ACC102	SLA_FIRST	General	2	23					########	########	15	General w	Depot_Arli	1
WO00000	Synthetic c	32.94634	-96.7028	ACC066	SLA_FIRST	Electrical	2	33				Electrical	########	########	15	Electrical v	Depot_Ricl	1
WO00000	Synthetic c	32.94114	-96.7138	ACC063	SKILL_FIRS	Electrical	2	36				Electrical	########	########	45	Electrical v	Depot_Ricl	0
WO00000	Synthetic c	32.94568	-96.7665	ACC050	SLA_FIRST	Electrical	1	40				Electrical	########	########	15	Electrical v	Depot_Ricl	1
WO00000	Synthetic c	33.00671	-96.6903	ACC031	SKILL_FIRS	General	2	40					########	########	45	General w	Depot_Plar	0
WO00001	Synthetic c	33.11378	-96.7969	ACC031	SKILL_FIRS	HVAC	2	58				HVAC	########	########	45	HVAC worl	Depot_Fris	0
WO00001	Synthetic c	32.79423	-97.0458	ACC117	TRAVEL_FI	Controls	2	58				Controls	########	########	45	Controls w	Depot_Arli	0
WO00001	Synthetic c	33.15903	-96.8242	ACC117	TRAVEL_FI	Controls	2	37				Controls	########	########	45	Controls w	Depot_Fris	0
WO00001	Synthetic c	33.16408	-96.6291	ACC036	OVERTIME	Controls	2	58				Controls	########	########	45	Controls w	Depot_Mcl	0
WO00001	Synthetic c	32.95525	-96.738	ACC066	SLA_FIRST	Controls	2	47				Controls	########	########	15	Controls w	Depot_Ricl	1
WO00001	Synthetic c	32.84738	-96.9413	ACC064	OVERTIME	Controls	2	61				Controls	########	########	45	Controls w	Depot_Las	0
WO00001	Synthetic c	33.16522	-96.825	ACC089	TRAVEL_FI	General	3	33					########	########	45	General w	Depot_Fris	0
WO00001	Synthetic c	33.16734	-96.7965	ACC099	SKILL_FIRS	General	1	33					########	########	45	General w	Depot_Fris	0
WO00001	Synthetic c	33.03851	-96.7355	ACC109	TRAVEL_FI	Electrical	3	33				Electrical	########	########	45	Electrical v	Depot_Plar	0
WO00001	Synthetic c	33.0377	-96.7133	ACC081	TRAVEL_FI	HVAC	2	55				HVAC	########	########	45	HVAC worl	Depot_Plar	0
WO00002	Synthetic c	32.957	-96.7411	ACC093	TRAVEL_FI	General	3	27					########	########	45	General w	Depot_Ricl	0
WO00002	Synthetic c	33.19502	-96.5916	ACC091	SKILL_FIRS	Plumbing	1	49				Plumbing	########	########	45	Plumbing v	Depot_Mcl	0
WO00002	Synthetic c	33.05238	-96.7183	ACC025	TRAVEL_FI	General	3	28					########	########	45	General w	Depot_Plar	0
WO00002	Synthetic c	33.13964	-96.7872	ACC101	TRAVEL_FI	Controls	2	62				Controls	########	########	45	Controls w	Depot_Fris	0
WO00002	Synthetic c	33.16937	-96.6288	ACC113	TRAVEL_FI	Electrical	2	50				Electrical	########	########	45	Electrical v	Depot_Mcl	0
WO00002	Synthetic c	32.94716	-96.8472	ACC002	SLA_FIRST	Controls	1	43				Controls	########	########	15	Controls w	Depot_Dall	1
WO00002	Synthetic c	32.90902	-96.644	ACC096	OVERTIME	Plumbing	2	46				Plumbing	########	########	45	Plumbing v	Depot_Gar	0

Routes

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
ObjectID	Name	StartDepot	EndDepot	EarliestSta	LatestStart	Capacities	CostPerUr	CostPerUr	CostPerUr	OvertimeS	MaxOrder(	MaxTotalTi	Assignmer	SpecialtyN	Descriptio	Technician	Team	
1	Technician	Depot_Ricl	Depot_Ricl	########	########	150	1	0.35	4.5	540	130	720	1	Electrical	Synthetic s	4.52	Richardson	
2	Technician	Depot_Dall	Depot_Dall	########	########	150	1	0.35	4.5	540	130	720	1	HVAC	Synthetic s	3.95	Dallas_North	
3	Technician	Depot_Las	Depot_Las	########	########	150	1	0.35	4.5	540	130	720	1	Controls	Synthetic s	4.66	Las_Colinas	
4	Technician	Depot_Plar	Depot_Plar	########	########	150	1	0.35	4.5	540	130	720	1	Plumbing E	Synthetic s	4.39	Plano	
5	Technician	Depot_Fris	Depot_Fris	########	########	150	1	0.35	4.5	540	130	720	1	General	Synthetic s	3.36	Frisco	
6	Technician	Depot_Arli	Depot_Arli	########	########	150	1	0.35	4.5	540	130	720	1	Electrical	Synthetic s	4.86	Arlington	
7	Technician	Depot_Gar	Depot_Gar	########	########	150	1	0.35	4.5	540	130	720	1	HVAC	Synthetic s	4.49	Garland	
8	Technician	Depot_Mcl	Depot_Mcl	########	########	150	1	0.35	4.5	540	130	720	1	Controls G	Synthetic s	4.54	McKinney	
9	Technician	Depot_Ricl	Depot_Ricl	########	########	150	1	0.35	4.5	540	130	720	1	Plumbing	Synthetic s	3.42	Richardson	
10	Technician	Depot_Dall	Depot_Dall	########	########	150	1	0.35	4.5	540	130	720	1	General	Synthetic s	3.97	Dallas_North	
11	Technician	Depot_Las	Depot_Las	########	########	150	1	0.35	4.5	540	130	720	1	Electrical	Synthetic s	3.83	Las_Colinas	
12	Technician	Depot_Plar	Depot_Plar	########	########	150	1	0.35	4.5	540	130	720	1	HVAC Plur	Synthetic s	4.78	Plano	
13	Technician	Depot_Fris	Depot_Fris	########	########	150	1	0.35	4.5	540	130	720	1	Controls	Synthetic s	4.29	Frisco	
14	Technician	Depot_Arli	Depot_Arli	########	########	150	1	0.35	4.5	540	130	720	1	Plumbing	Synthetic s	4.6	Arlington	
15	Technician	Depot_Gar	Depot_Gar	########	########	150	1	0.35	4.5	540	130	720	1	General	Synthetic s	3.95	Garland	
16	Technician	Depot_Mcl	Depot_Mcl	########	########	150	1	0.35	4.5	540	130	720	1	Electrical (	Synthetic s	3.59	McKinney	
17	Technician	Depot_Ricl	Depot_Ricl	########	########	150	1	0.35	4.5	540	130	720	1	HVAC	Synthetic s	4.14	Richardson	
18	Technician	Depot_Dall	Depot_Dall	########	########	150	1	0.35	4.5	540	130	720	1	Controls	Synthetic s	3.31	Dallas_North	
19	Technician	Depot_Las	Depot_Las	########	########	150	1	0.35	4.5	540	130	720	1	Plumbing	Synthetic s	4.61	Las_Colinas	
20	Technician	Depot_Plar	Depot_Plar	########	########	150	1	0.35	4.5	540	130	720	1	General H	Synthetic s	4.27	Plano	
21	Technician	Depot_Fris	Depot_Fris	########	########	150	1	0.35	4.5	540	130	720	1	Electrical	Synthetic s	4.49	Frisco	
22	Technician	Depot_Arli	Depot_Arli	########	########	150	1	0.35	4.5	540	130	720	1	HVAC	Synthetic s	3.8	Arlington	
23	Technician	Depot_Gar	Depot_Gar	########	########	150	1	0.35	4.5	540	130	720	1	Controls	Synthetic s	4.85	Garland	
24	Technician	Depot_Mcl	Depot_Mcl	########	########	150	1	0.35	4.5	540	130	720	1	Plumbing E	Synthetic s	4.72	McKinney	
25	Technician	Depot_Ricl	Depot_Ricl	########	########	150	1	0.35	4.5	540	130	720	1	General	Synthetic s	4.52	Richardson	
26	Technician	Depot_Dall	Depot_Dall	########	########	150	1	0.35	4.5	540	130	720	1	Electrical	Synthetic s	3.53	Dallas_North	

Depots

A	B	C	D
Name	Latitude	Longitude	
Depot_Richardson	32.9483	-96.7299	
Depot_Dallas_North	32.9265	-96.8041	
Depot_Las_Colinas	32.8725	-96.9414	
Depot_Plano	33.0198	-96.6989	
Depot_Frisco	33.1507	-96.8236	
Depot_Arlington	32.7357	-97.1081	
Depot_Garland	32.9126	-96.6389	
Depot_McKinney	33.1976	-96.6153	

Daily timeline (nightly baseline → live updates)

- **T0** Base reference scenario
- **T1** EAS 1 Start variance
- **T2** EAS 2 Finish variance
- **T3** EAS 3 Skipped stop
- **T4** EAS 4 Auto re-plan
- **T5** RT 1 Urgent activity
- **T6** RT 2 Cancellation
- **T7** RT 3 Duration change
- **T8** RT 4 Shift change
- **T9** RT 5 Location hours
- **T10** RT 6 Technician sick
- **T11** RT 7 Combined live flow
- **T12** GB 1 Travel first
- **T13** GB 2 Quality / specialty
- **T14** GB 3 SLA first
- **T15** GB 4 Reduce overtime
- **T16** GB 5 Customer weights

Summary of the PoC input data and workflow

The same sample-deck narrative is kept, but the underlying demonstration universe is now the recorded 150 work-order / 50-route model.

Demo orders

150 work orders
customer / location / timing / SLA fields

Demo routes

50 routes / technicians
shift windows, capacities, overtime, specialties

Demo depot

1 depot
regional start / end anchor for the routes

accounts + goal_profiles

customer-level weighting
travel, SLA, rating, and overtime

Workflow from data to scenario maps

1. Load and validate the recorded 150-work-order / 50-route inputs.
2. Convert tables to ArcGIS features and solve the reference VRP.
3. Re-run the solve after each targeted event or goal-weight change.
4. Preserve execution times from the recorded scenario log.
5. Present every requirement in the same screenshot-style card layout.

Representative GIS input map

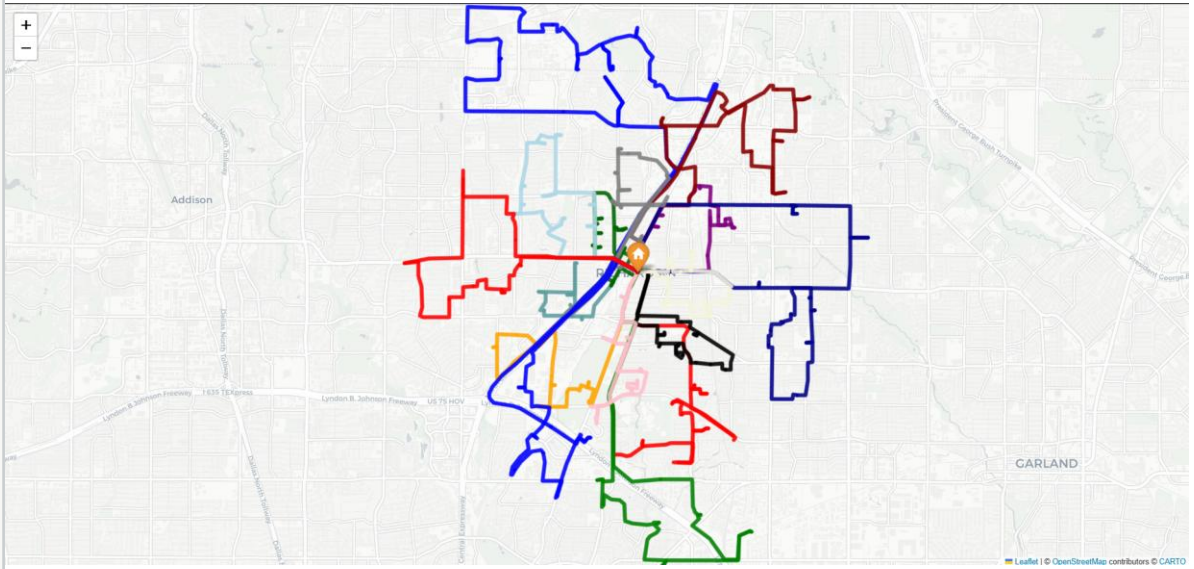
150 work orders · 50 routes · 1 depot

Orders, route geometries, and recorded execution times are carried into the deck for comparison.

Baseline reference

Reference plan used as the comparison point for the 16 Sid requirement scenarios.

Original scenario map



Baseline reference

Base scenario

50 active routes

150 served orders

Input changed	Routing effect	Execution time
None	Reference plan	40.8 sec

What changed in the notebook

Nothing was modified. The notebook loads the recorded 150-work-order / 50-route inputs, builds the impacted visualization subset, and solves the base VRP before any requirement event is triggered.

Observed output / interpretation

This reference subset becomes the benchmark for all later comparisons. It keeps the map readable while still representing the recorded 150-work-order / 50-route run.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

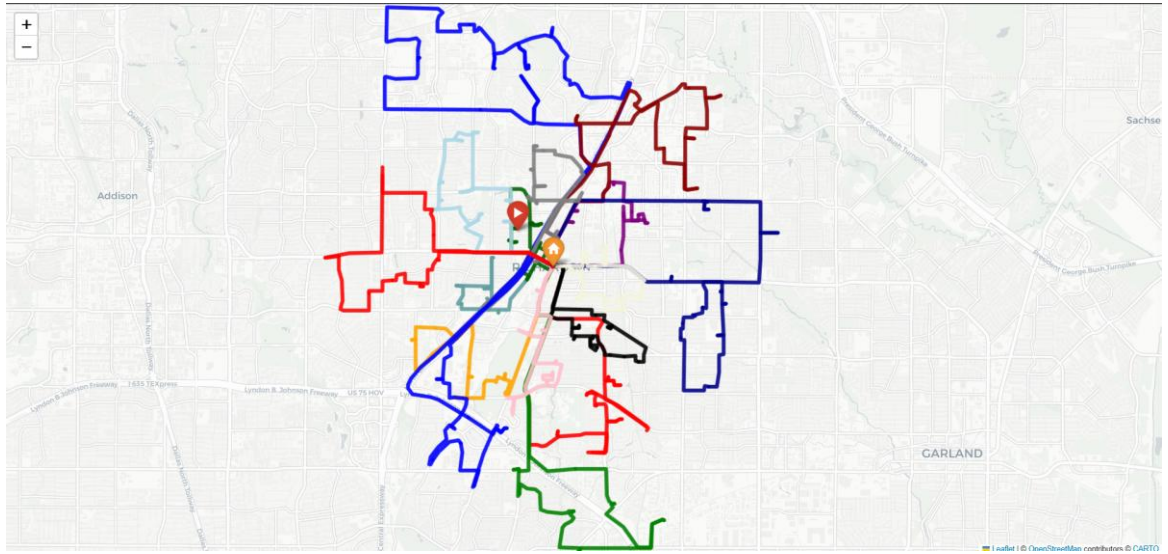
Reference subset ready for comparison across all later scenarios.

Metric	Base	Now	Note
Orders	n/a	150	source
Routes	n/a	50	source
Exec time	n/a	40.8 sec	T0 logged

Jobs starting earlier or later than planned

Notebook walkthrough using the original map output for EAS 1.

Original scenario map



Jobs starting earlier or later than planned

Scenario 1 - Start variance

Technician start shifted | **+20 min**

Input changed	Routing effect	Execution time
Routes	Timing shift	34.4 sec

What changed in the notebook

Adjust the impacted technician's EarliestStartTime and LatestStartTime by the live variance, then re-solve the impacted subset.

Observed output / interpretation

The subset can keep the same coverage while the impacted route starts later. This is the execution-aware equivalent of the sample traffic-delay slide.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

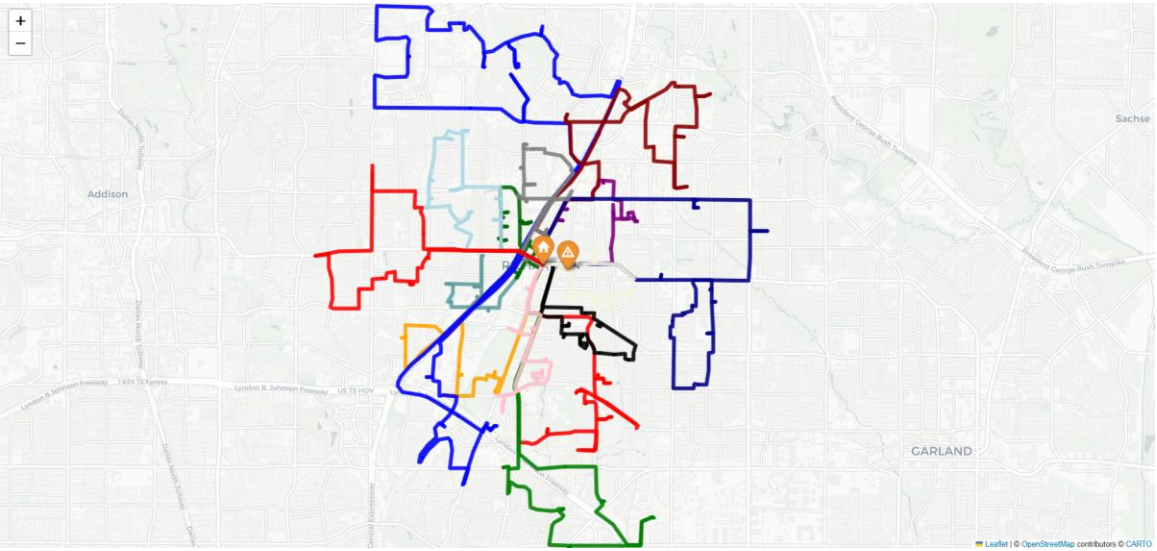
Impacted technician shifts later while subset coverage stays stable.

Metric	Base	Now	Note
Route start	07:00	07:20	+20 min
Routes	50	50	stable
Orders	150	150	served

Jobs finishing earlier or later

Notebook walkthrough using the original map output for EAS 2.

Original scenario map



Jobs finishing earlier or later

Scenario 2 - Finish variance

Service duration changed

+25 min

Input changed	Routing effect	Execution time
Orders	Downstream timing	38.3 sec

What changed in the notebook

Update the active order's ServiceTime using the finish variance and re-solve the impacted subset so downstream activities reflect the new duration.

Observed output / interpretation

The key business effect is compression or expansion of downstream slack around the changed stop, not a full-universe re-plan.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

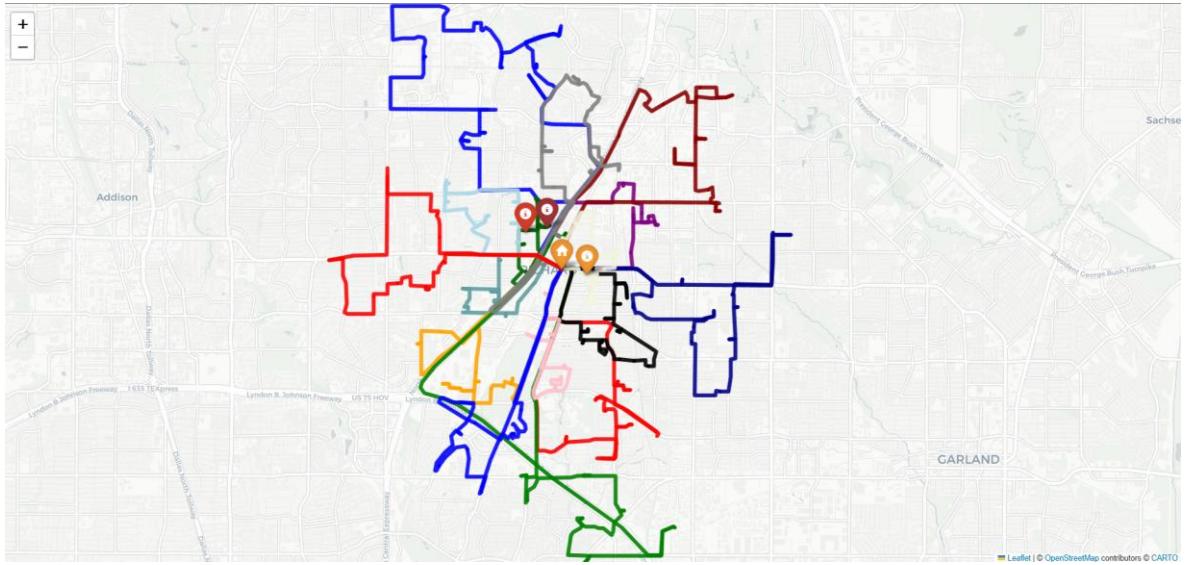
Finish variance changes local slack around the impacted order.

Metric	Base	Now	Note
ServiceTime	30 min	55 min	edited
Routes	50	50	stable
Orders	150	150	served

Technicians skipping scheduled job order

Notebook walkthrough using the original map output for EAS 3.

Original scenario map



Technicians skipping scheduled job order

Scenario 3 - Skipped stop

Skipped stop reinserted | **Priority raised**

Input changed	Routing effect	Execution time
Orders	Sequence repair	34.2 sec

What changed in the notebook

Clear any sequence bias for the skipped job, elevate its operational priority, and let re-optimization place it back into the best feasible route sequence.

Observed output / interpretation

The order stays in the problem, but the old sequence is no longer trusted. This is the right execution-aware response to a skipped stop.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

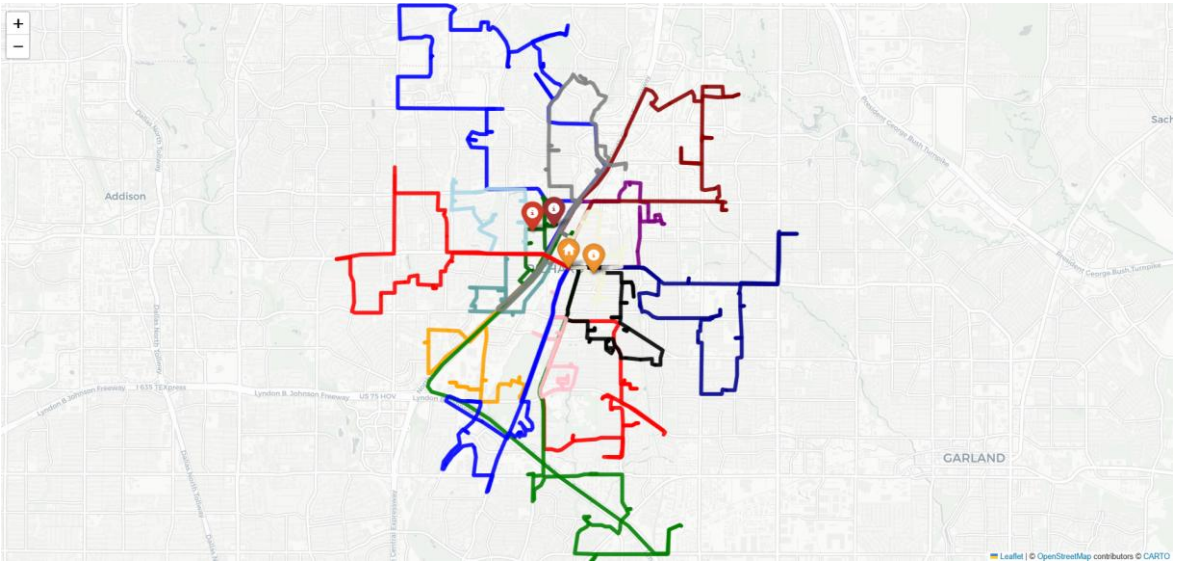
Skipped work is reinserted for repair instead of being silently dropped.

Metric	Base	Now	Note
Skipped order	fixed seq.	free	repair
Priority	normal	high	elevated
Coverage	150	150	served

Automatically re-optimize downstream activities

Notebook walkthrough using the original map output for EAS 4.

Original scenario map



Automatically re-optimize downstream activities

Scenario 4 - Auto re-plan

3 live signals combined | **1 rolling solve**

Input changed	Routing effect	Execution time
Orders + routes	Combined replan	34.1 sec

What changed in the notebook

Queue the start variance, finish variance, and skipped-job signals together, then run one impacted-subset re-solve so downstream assignments refresh in one pass.

Observed output / interpretation

This is the end-state execution-aware design Sid asked for: automatic downstream repair instead of one-off manual comparison slides.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

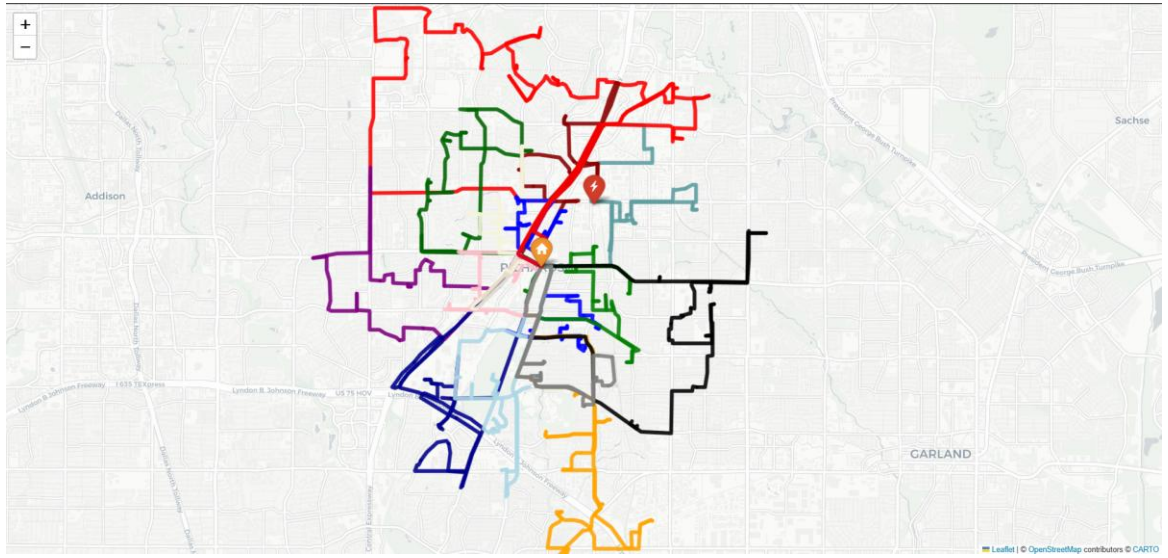
Automatic downstream repair combines multiple execution signals in one pass.

Metric	Base	Now	Note
Signals	1	3	combined
Solve mode	manual	rolling	auto
Orders	150	150	served

New or urgent activities

Notebook walkthrough using the original map output for RT 1.

Original scenario map



New or urgent activities

Scenario 5 - Urgent activity

Urgent job inserted | **+1 order**

Input changed	Routing effect	Execution time
Orders	Fast insertion	42.3 sec

What changed in the notebook

Add a new high-priority order with a tight time window, specialty, preferred depot, and SLA-first profile, then re-solve the impacted subset.

Observed output / interpretation

The urgent job enters the schedule without a full demo-scope re-solve. That is the core real-time behavior Sid wanted.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	151	served

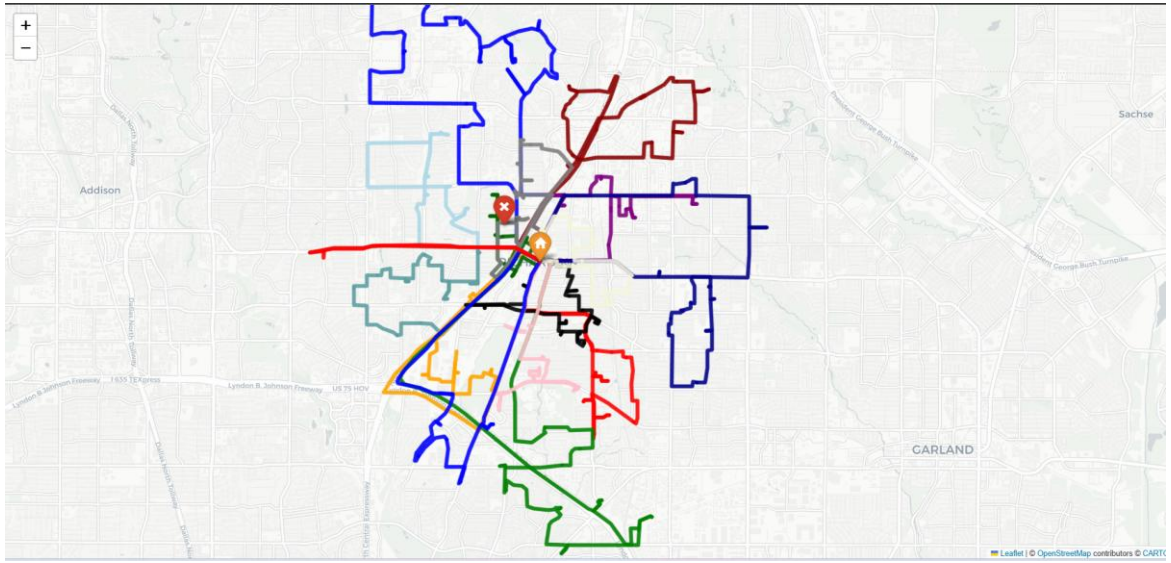
Urgent activity is absorbed quickly while route count stays stable.

Metric	Base	Now	Note
Orders	150	151	served
Routes	50	50	active
Urgent WO	none	inserted	tight TW

Cancellations

Notebook walkthrough using the original map output for RT 2.

Original scenario map



Cancellations

Scenario 6 - Cancellation

1 order removed

Demand reduced

Input changed	Routing effect	Execution time
Orders	Demand reduction	40.1 sec

What changed in the notebook

Remove the cancelled order from orders_df and re-solve the impacted subset so route time and sequence compress around the updated demand.

Observed output / interpretation

Capacity is released immediately and the remaining work can rebalance. This mirrors the sample cancellation slide, but at the 150-order / 50-route demo scale.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	149	served

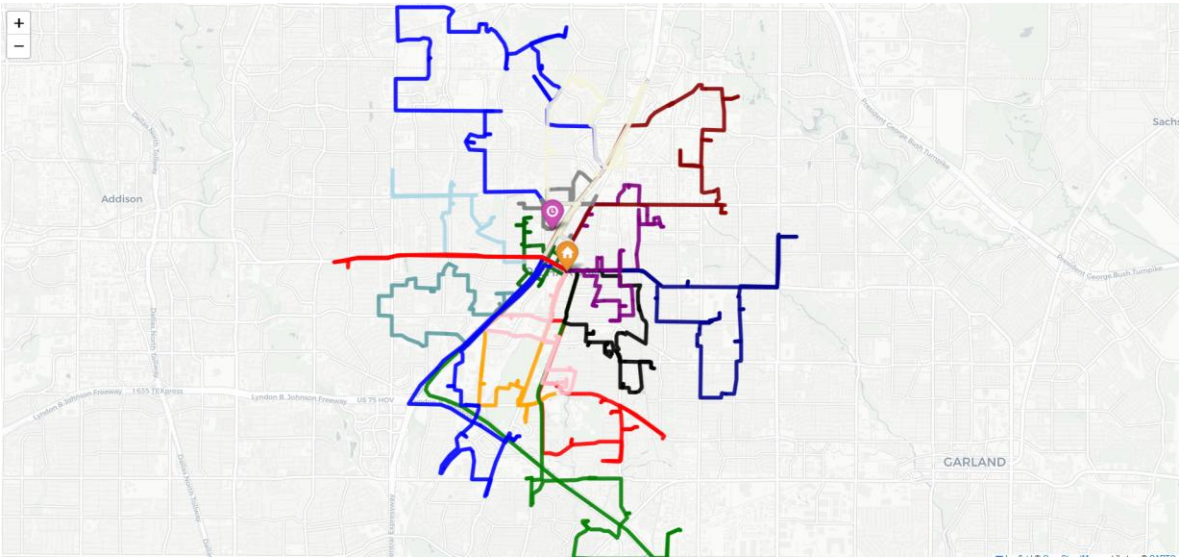
Cancellation frees capacity and reduces served demand in the subset.

Metric	Base	Now	Note
Orders	150	149	served
Routes	50	50	active
Cancelled	0	1	removed

Duration changes

Notebook walkthrough using the original map output for RT 3.

Original scenario map



Duration changes

Scenario 7 - Duration change

Activity duration updated

95 min example

Input changed	Routing effect	Execution time
Orders	Local timing shift	35.1 sec

What changed in the notebook

Update the affected order's ServiceTime and re-solve the impacted subset to rebalance timing around the changed stop.

Observed output / interpretation

Only the relevant order needs to change. The route timing around that location expands or contracts accordingly.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

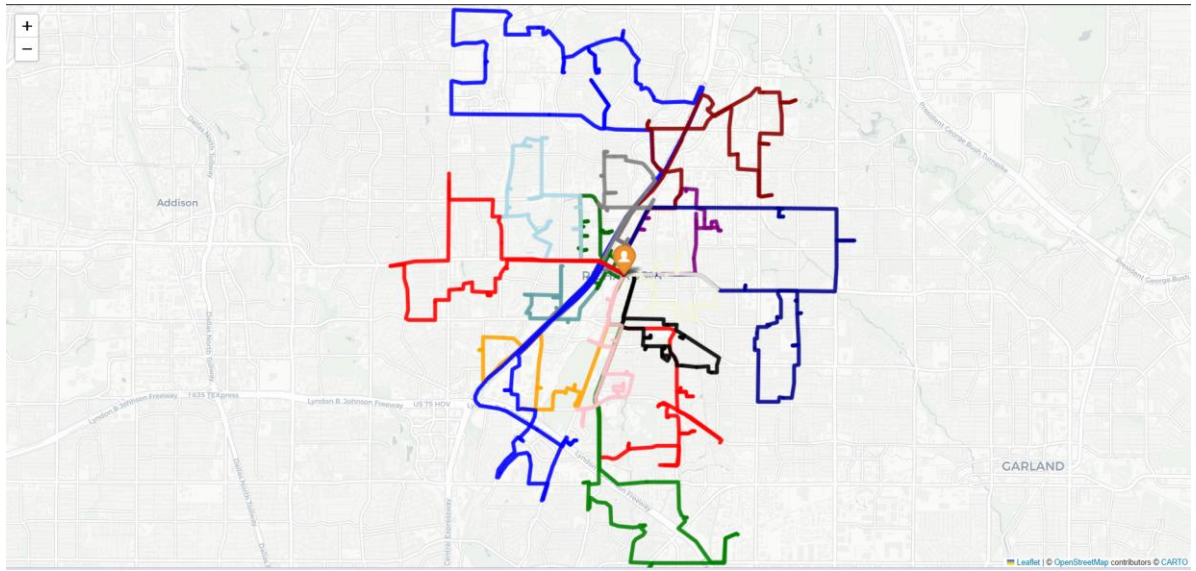
A local duration edit changes nearby timing without a full-universe rebuild.

Metric	Base	Now	Note
ServiceTime	30 min	95 min	edited
Routes	50	50	stable
Orders	150	150	served

Technician availability or shift changes

Notebook walkthrough using the original map output for RT 4.

Original scenario map



Technician availability or shift changes

Scenario 8 - Shift change

Shift window changed | **+30 / +45**

Input changed	Routing effect	Execution time
Routes	Capacity rebalance	37.8 sec

What changed in the notebook

Shift the technician's start and end window and re-solve the impacted subset so work can move away from the constrained route if required.

Observed output / interpretation

This is more nuanced than removing a route entirely. The technician still exists, but their usable capacity changes in real time.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

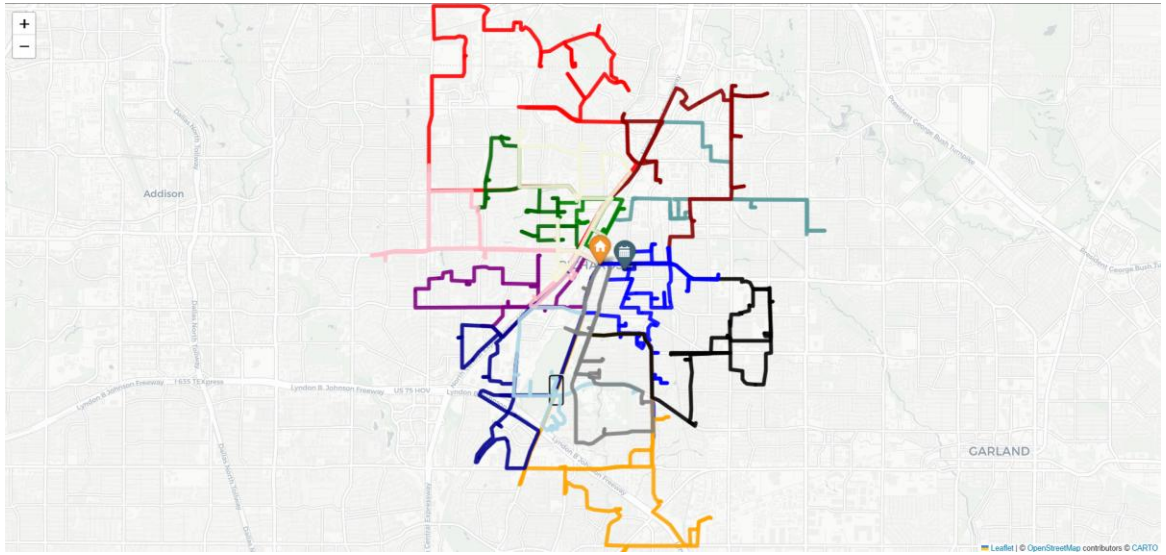
Shift edits keep the technician in the model while changing usable capacity.

Metric	Base	Now	Note
Start shift	07:00	07:30	later
End shift	16:00	16:45	later
Rebalance	none	possible	subset

Location hours changes

Notebook walkthrough using the original map output for RT 5.

Original scenario map



Location hours changes

Scenario 9 - Location hours

Time window tightened | **04:00-08:00**

Input changed	Routing effect	Execution time
Orders	Window constraint	40.5 sec

What changed in the notebook

Edit the order's time window and re-solve the impacted subset using the updated customer-availability constraint.

Observed output / interpretation

This shows that real-time scheduling is not only about technician status; customer-side availability can also force route resequencing or reassignment.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

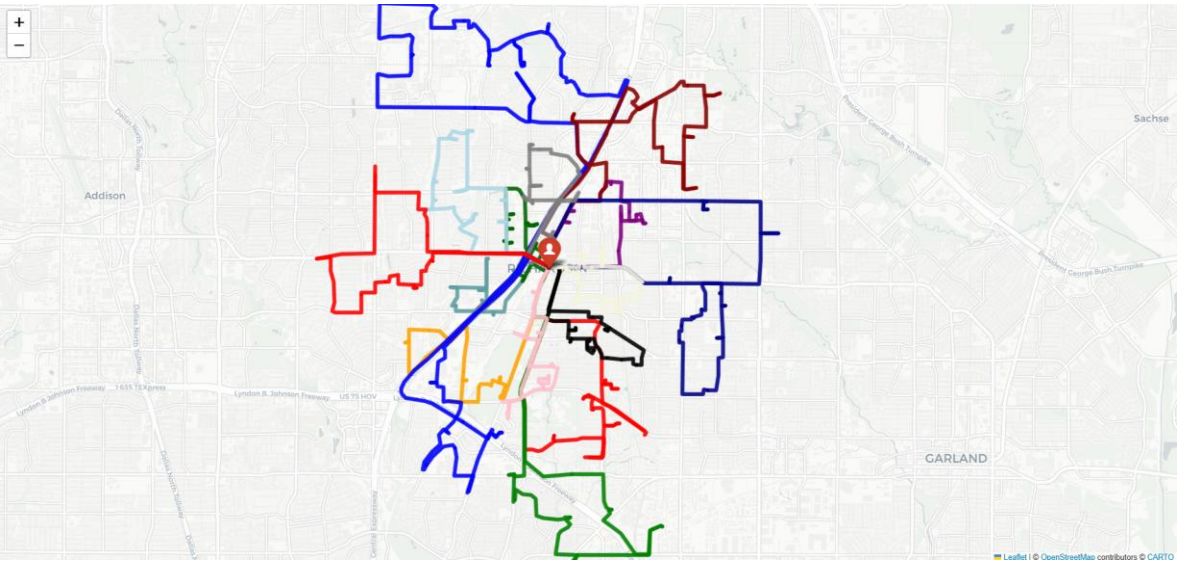
Customer-side availability changes can force automatic resequencing.

Metric	Base	Now	Note
TW start	open	04:00	edited
TW end	open	08:00	edited
Effect	none	reseq.	window

Technician calls in sick

Notebook walkthrough using the original map output for RT 6.

Original scenario map



Technician calls in sick

Scenario 10 - Tech sick

1 technician removed | **Fleet reduced**

Input changed	Routing effect	Execution time
Routes	Fleet reduction	41.4 sec

What changed in the notebook

Drop the unavailable route from routes_df and re-solve the impacted subset using the reduced fleet.

Observed output / interpretation

This is the highest-disruption real-time scenario. The business view is either rebalanced coverage across the remaining routes or a partial solution when the subset is too constrained.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	49	49	active
Orders	150	150	served

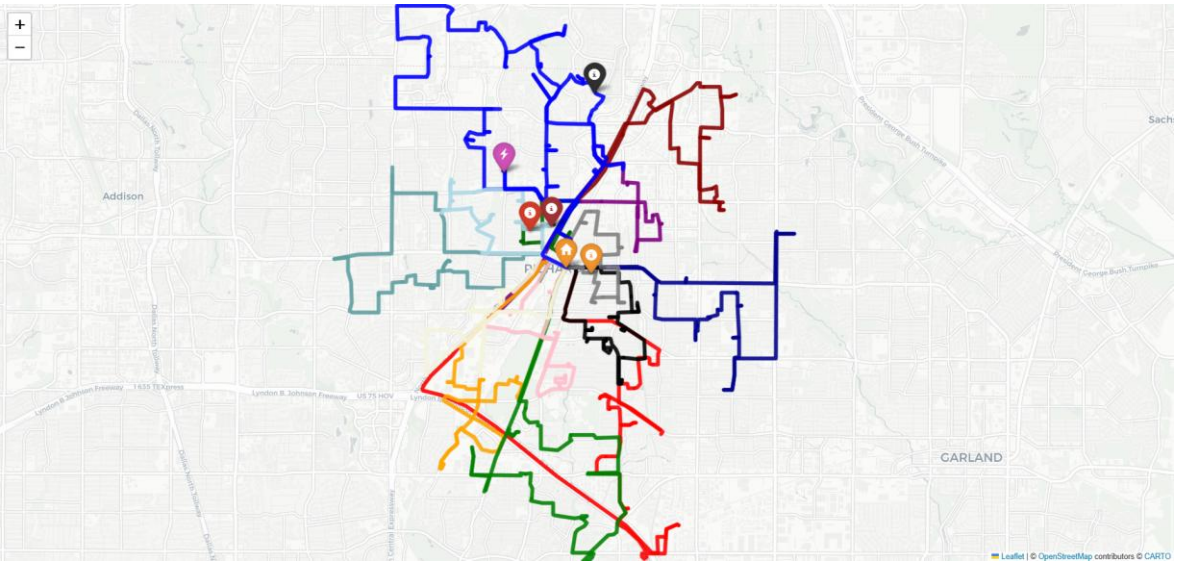
One less route increases disruption and can expose unrouted work.

Metric	Base	Now	Note
Routes	50	49	active
Coverage	150	partial	possible
Impact	normal	high	severe

All EAS scenarios included in real-time flow

Notebook walkthrough using the original map output for RT 7.

Original scenario map



All EAS scenarios included in real-time flow

Scenario 11 - Combined flow

8 triggers combined

1 rolling solve

Input changed	Routing effect	Execution time
Event queue	Rolling replan	36.9 sec

What changed in the notebook

Create one event list containing execution-aware and real-time triggers, then run a single impacted-subset re-optimization.

Observed output / interpretation

This is the end-state story for Sid: one near-real-time event pipeline, one impacted-subset solve, and one continuously updated route picture.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

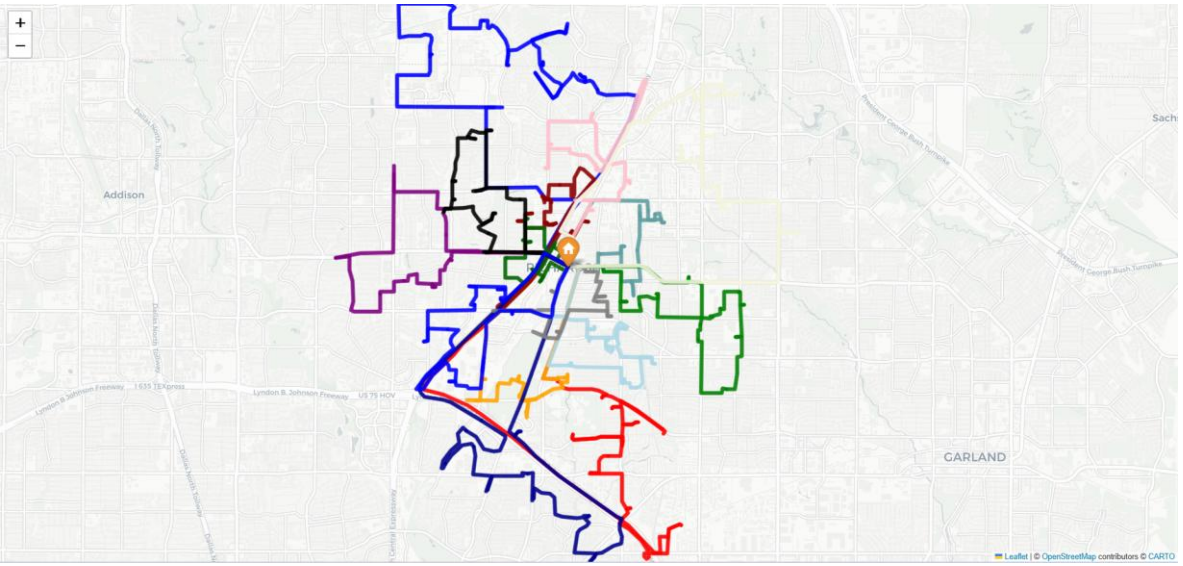
A single rolling pipeline orchestrates all live event classes together.

Metric	Base	Now	Note
Event types	1	8	combined
Solve mode	manual	rolling	live
Orders	150	150	served

Minimize travel time

Notebook walkthrough using the original map output for GB 1.

Original scenario map



Minimize travel time

Scenario 12 - Travel first

Travel weighted highest | **Goal profile**

Input changed	Routing effect	Execution time
Goal weights	Travel minimization	41.8 sec

What changed in the notebook

Increase the travel-time weight in the route cost structure so the solver prefers more compact route patterns when multiple solutions are feasible.

Observed output / interpretation

This is the cleanest goal-based benchmark: if travel is weighted highest, the route layout should become more geographically compact.

Reference subset snapshot used as the comparison point.

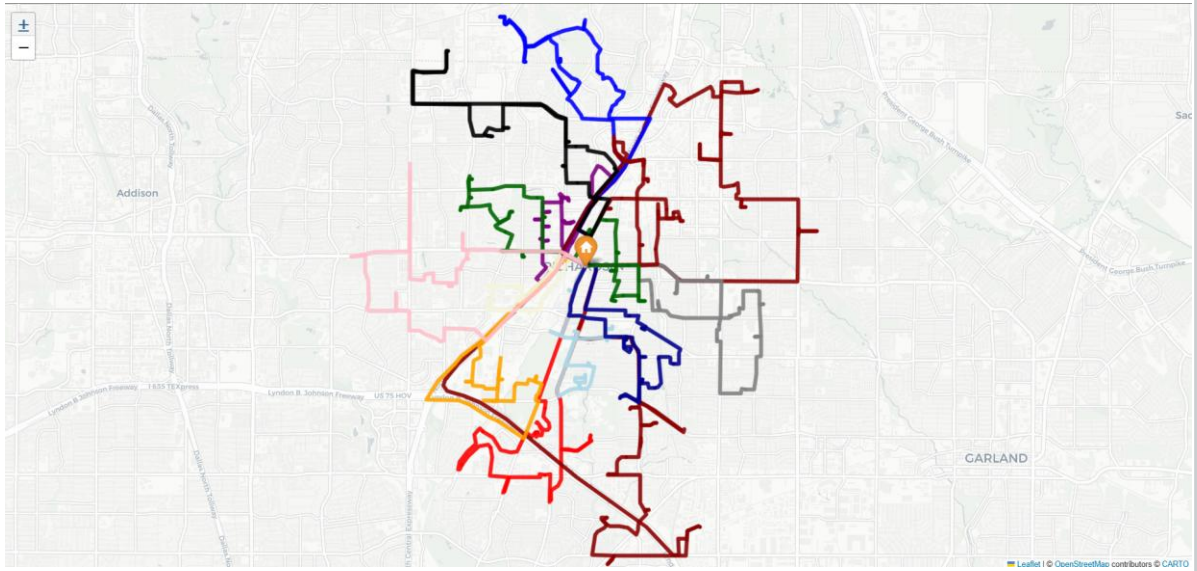
Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

The optimization target explicitly favors shorter travel and tighter clusters.

Metric	Base	Now	Note
Travel weight	base	high	goal
Pattern	neutral	compact	expected
Coverage	150	150	served

Prefer higher-rated or specialized technicians

Notebook walkthrough using the original map output for GB 2.



Scenario 13 - Quality / specialty

Technician bias added

Skill matched

Input changed	Routing effect	Execution time
Weights + skills	Skill-based bias	40.4 / 52.5 sec

What changed in the notebook

Blend technician-rating bias with specialty constraints so premium or typed work is steered toward the technicians best matched to deliver it.

Observed output / interpretation

This single business requirement is represented by two notebook patterns: one for higher-rated assignment bias and one for specialty-constrained assignment.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

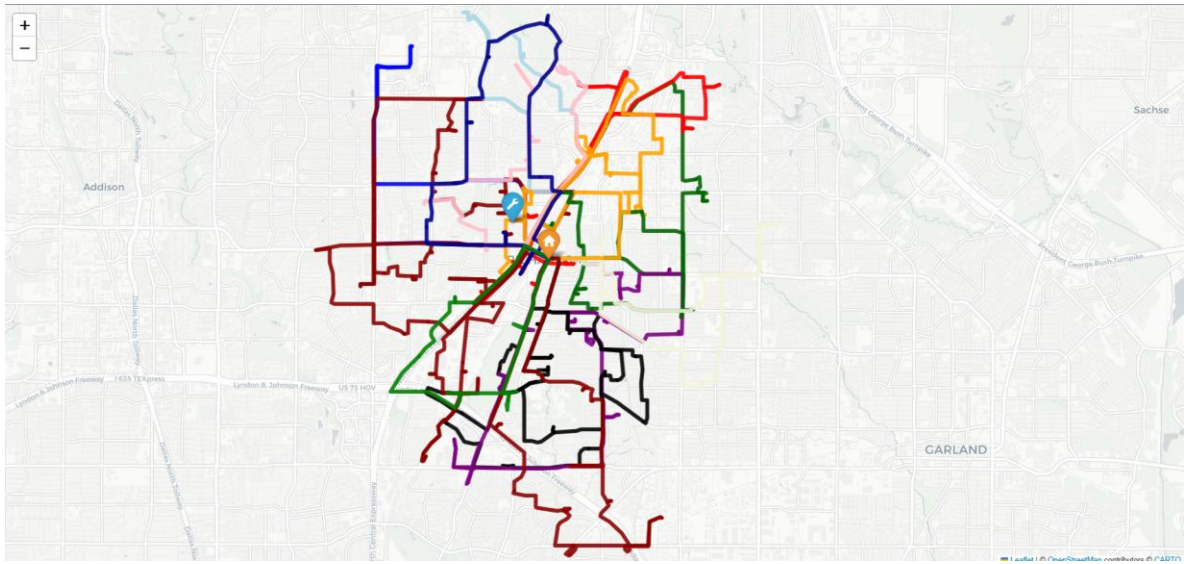
Quality and specialty preferences make the route plan more technician-aware.

Metric	Base	Now	Note
Rating bias	off	on	weighted
Specialty	soft	matched	typed
Coverage	150	150	served

Maximize on-time completion / SLA adherence

Notebook walkthrough using the original map output for GB 3.

Original scenario map



Maximize on-time completion / SLA adherence

Scenario 14 - SLA first

SLA-sensitive work protected

Violation tightened

Input changed	Routing effect	Execution time
Weights + SLA	Priority to SLA	49.4 sec

What changed in the notebook

Apply account goal profiles, tighten time-window violation tolerance for SLA-sensitive orders, and re-solve so these jobs are protected first.

Observed output / interpretation

This is the clearest enterprise-value story: optimal is not always the shortest route; sometimes it is the route that protects SLA commitments first.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

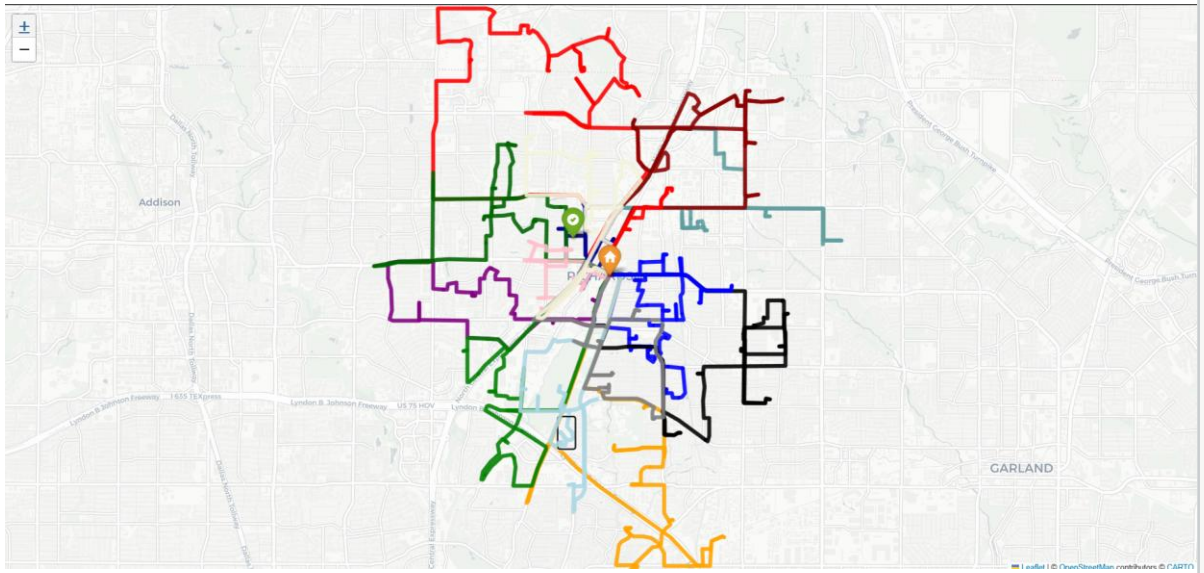
SLA adherence becomes the dominant optimization target for sensitive work.

Metric	Base	Now	Note
SLA list	none	targeted	orders
Violation	base	tight	tolerance
Goal	neutral	SLA first	weighted

Reduce overtime

Notebook walkthrough using the original map output for GB 4.

Original scenario map



Reduce overtime

Scenario 15 - Reduce overtime

Overtime penalty raised | **Regular time first**

Input changed	Routing effect	Execution time
Route cost	Lower OT usage	41.5 sec

What changed in the notebook

Raise the overtime penalty and move the overtime threshold earlier so the solver uses regular-time capacity first whenever it can.

Observed output / interpretation

The expected route behavior is more balanced regular-time utilization, with overtime becoming a deliberate exception instead of the default outcome.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

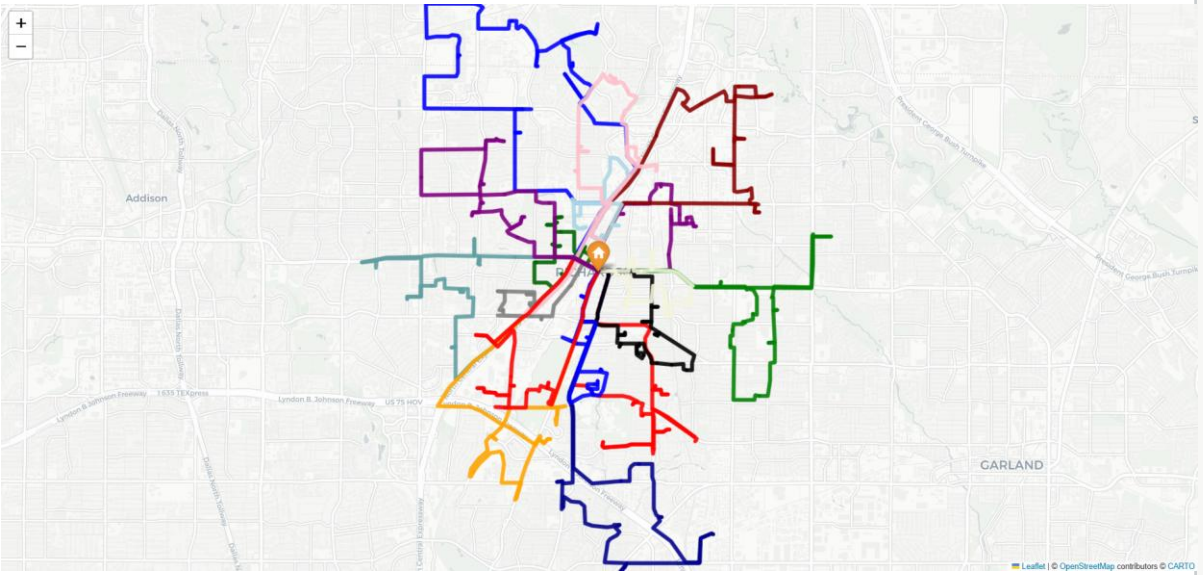
Overtime becomes a penalized exception instead of the default operating mode.

Metric	Base	Now	Note
OT threshold	base	earlier	edited
OT penalty	base	high	weighted
Effect	neutral	lower OT	expected

Customer-defined soft-constraint weights

Notebook walkthrough using the original map output for GB 5.

Original scenario map



Customer-defined soft-constraint weights

Scenario 16 - Customer weights

Account profile applied | **Custom optimum**

Input changed	Routing effect	Execution time
Account profiles	Customer-specific optimum	42.7 sec

What changed in the notebook

Apply account-level goal profiles and a blended customer weight set across travel, rating, SLA adherence, and overtime, then solve to that customer-specific policy.

Observed output / interpretation

This turns the VRP engine into a configurable optimization service rather than a single hard-coded routing policy.

Reference subset snapshot used as the comparison point.

Metric	Base	Now	Note
Routes	50	50	active
Orders	150	150	served

Each customer can define its own notion of optimal through weighted soft constraints.

Metric	Base	Now	Note
Travel	base	weighted	profile
SLA / rating	base	weighted	profile
Overtime	base	weighted	profile

Summary - Base plan vs all 16 scenarios (1/2)

Visual summary of how each scenario affects route timing, service coverage, and business impact.

1. Start variance

Routes	Served	Key
50	150	07:20

2. Finish variance

Routes	Served	Key
50	150	55 min

3. Skipped stop

Routes	Served	Key
50	150	Repair

4. Auto re-plan

Routes	Served	Key
50	150	3 signals

5. Urgent activity

Routes	Served	Key
50	151	+1 WO

6. Cancellation

Routes	Served	Key
50	149	-1 WO

7. Duration change

Routes	Served	Key
50	150	95 min

8. Shift change

Routes	Served	Key
50	150	+30/+45

Biggest timing shift
Start variance (+20 min)

Biggest demand change
Urgent +1 / Cancel -1

Most workflow change
Auto re-plan (3 signals)

Detailed comparison

Scenario	What changed vs Base	Routes	Orders / Routes	Impacted timing	Operational impact
Start variance	Tech delayed +20 min	15	150 / 50	07:20 start	Coverage stable
Finish variance	Service time +25 min	15	150 / 50	Local slack	Downstream adjusts
Skipped stop	Skipped job reinserted	15	150 / 50	Seq repair	Missed work retained
Auto re-plan	3 live signals combined	15	150 / 50	Rolling repair	Auto downstream update
Urgent activity	Urgent order inserted	15	151 / 50	Route extends	Urgent work absorbed
Cancellation	1 active order removed	15	149 / 50	Route compresses	Capacity rebalanced
Duration change	ServiceTime edited live	15	150 / 50	Nearby timings	Local resequencing
Shift change	Shift window shifted	15	150 / 50	Usable capacity	Work can rebalance

Quick comparison to base

Scenario	Comparison to Base	Base subset	Result
Start variance	Same coverage as base	50 routes / 150 served	Route starts later by 20 min
Finish variance	Same coverage as base	50 routes / 150 served	Slack expands at impacted stop
Skipped stop	Same coverage	50 routes / 150 served	Skipped order re-optimized
Auto re-plan	More dynamic	50 routes / 150 served	One rolling solve refreshes plan
Urgent activity	+1 order vs base	50 routes / 150 served	Subset absorbs urgent work
Cancellation	-1 order vs base	50 routes / 150 served	Freed capacity redistributed
Duration change	Same coverage	50 routes / 150 served	Only local timing changes
Shift change	Route still active	50 routes / 150 served	Capacity changes without removal

Summary - Base plan vs all 16 scenarios (2/2)

Visual summary of how each scenario affects route timing, service coverage, and business impact.

9. Location hours

Routes	Served	Key
50	150	04-08

10. Technician sick

Routes	Served	Key
49	Partial	-1 route

11. Combined flow

Routes	Served	Key
50	150	8 triggers

12. Travel first

Routes	Served	Key
50	150	Compact

13. Quality / specialty

Routes	Served	Key
50	150	Skill fit

14. SLA first

Routes	Served	Key
50	150	Protected

15. Reduce overtime

Routes	Served	Key
50	150	Lower OT

16. Customer weights

Routes	Served	Key
50	150	Profile

Most severe
Technician sick

Most strategic
Customer weights

Highest protection
SLA first

Detailed comparison

Scenario	What changed vs Base	Routes	Orders / Routes	Impacted timing	Operational impact
Location hours	Customer window tightened	15	150 / 50	Window rule	Auto resequencing
Technician sick	1 tech removed	14	150 / 49	Unrouted risk	Highest disruption
Combined flow	All EAS + RT triggers combined	15	150 / 50	Rolling replan	Single live flow
Travel first	Travel weight highest	15	150 / 50	Compact pattern	Drive time minimized
Quality / specialty	Rating bias + skill match	15	150 / 50	Assignment bias	Best-fit techs favored
SLA first	SLA-sensitive work prioritized	15	150 / 50	Tighter tolerance	On-time work protected
Reduce overtime	OT penalty raised	15	150 / 50	Regular time first	Overtime discouraged
Customer weights	Account weights applied	15	150 / 50	Custom optimum	Each account defines optimal

Quick comparison to base

Scenario	Comparison to Base	Base subset	Result
Location hours	Customer-side change	150 served / 50 routes	Visit must respect new hours
Technician sick	-1 route	50 active routes	Remaining fleet may not absorb all work
Combined flow	One pipeline	50 routes / 150 served	One pipeline handles all events
Travel first	Target changed	50 routes / 150 served	Solver favors shorter legs
Quality / specialty	Tech-aware plan	50 routes / 150 served	Higher-rated / skilled techs preferred
SLA first	Business goal changed	50 routes / 150 served	SLA commitments dominate choice
Reduce overtime	Cost policy changed	50 routes / 150 served	Overtime becomes exception
Customer weights	Custom objective	50 routes / 150 served	VRP becomes configurable service